Smart Inventory Demand Forecasting Decision Support System

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***Abstract*—Inventory planning requires balancing product availability against the financial and operational costs associated with excess inventory. Conventional manual estimation and static replenishment rules may become inadequate when sales, pricing, inventory position, and supplier lead times vary simultaneously. This paper presents a modular web-based Smart Inventory Demand Forecasting Decision Support System (DSS) that integrates supervised machine learning, inventory-risk calculations, scenario analysis, approximate model interpretation, and a human-in-the-loop approval workflow.**

The implemented prototype was evaluated using a 48-observation chronological baseline dataset representing four products across twelve months. Five predictors were used to estimate demand: previous-period sales, inventory level, unit price, calendar month, and supplier lead time. Four supervised regression models—Linear Regression, Ridge Regression, Decision Tree, and Random Forest—were evaluated using a chronological 75:25 train-test split. On the 12-observation holdout set, Ridge Regression achieved the lowest Mean Absolute Error (MAE) of 13.39 and Mean Absolute Percentage Error (MAPE) of 1.70%, while Linear Regression achieved a slightly lower Root Mean Squared Error (RMSE) of 16.58 and higher R^2 of 0.9935 compared with 0.9926 for Ridge Regression. Ridge was therefore selected according to the predefined MAE-based model-selection criterion.

The reported results are treated as preliminary evidence because only 12 observations are available for final evaluation. The system additionally provides empirical uncertainty estimates, approximate feature-sensitivity explanations using baseline perturbation, inventory-risk indicators based on safety stock and reorder point calculations, and a scenario-analysis interface for what-if evaluation. A human-in-the-loop workflow allows users to approve, reject, or defer recommendations while maintaining an audit record. The contribution is positioned primarily as a systems-integration and decision-support contribution rather than as a new forecasting algorithm. The study demonstrates the feasibility of connecting demand prediction with operational inventory decisions while identifying the need for larger datasets, rolling-origin validation, calibrated uncertainty estimation, and stronger inventory-control evaluation before real-world deployment.

***Index Terms*—Demand forecasting, inventory management, decision support system, machine learning, Ridge regression, Random Forest, supply chain analytics, inventory optimization.**

I. INTRODUCTION

Accurate demand estimation is an important component of inventory planning because purchasing decisions must balance product availability against the financial and operational consequences of holding excess stock. Underestimating demand may result in stockouts, delayed fulfillment, and lost sales, whereas overestimating demand may increase holding costs and expose businesses to obsolete or slow-moving inventory. Recent research has therefore increasingly examined machine learning (ML) and artificial intelligence (AI) for demand forecasting within supply-chain and inventory-management environments [**?**], [9], [10].

Recent IEEE studies demonstrate the growing application of ML and deep learning to demand forecasting and inventory-related decision making. Poongothai *et al.* investigate machine learning for retail demand forecasting and inventory optimization [1]. Wang investigates deep-learning-based inventory optimization and intelligent demand prediction in e-commerce [2]. Zheng examines ML algorithms for enterprise supply-chain demand forecasting [3]. Singh *et al.* study deep-learning approaches for forecasting and inventory optimization in e-commerce supply chains [4]. These studies demonstrate that predictive models can provide useful information for supply-chain planning, but they also show the diversity of data requirements, algorithms, and operational contexts.

Other recent work focuses on more specific inventory and forecasting challenges. Sathyabama *et al.* integrate ARIMA and LSTM models with expiry prediction and dynamic pricing for retail inventory management [5]. Manos *et al.* present a dynamic demand-forecasting platform that combines machine learning with supply-chain value considerations [6]. Nurhasanah *et al.* investigate neural-network-based demand forecasting within an upstream and midstream supply-chain network [8]. Zuvieta *et al.* provide a recent IEEE Access

survey covering statistical and ML-based demand forecasting methods for inventory-oriented spare-parts environments [7].

Beyond IEEE publications, recent peer-reviewed research further emphasizes the importance of connecting forecasting with inventory decisions. Douaioui *et al.* provide a comprehensive review of ML and deep-learning models for supply-chain demand forecasting [9]. Sukolkit *et al.* explicitly connect demand forecasting with inventory management and replenishment decisions [11]. van der Haar *et al.* study supervised learning for integrated forecasting and inventory control [12]. Nguyen *et al.* integrate advanced time-series forecasting with probabilistic inventory decisions, including safety stock and reorder-point calculations [15]. An *et al.* investigate interpretable machine learning for demand forecasting in micro-fulfillment environments [13]. Liu *et al.* examine machine learning for inventory stockout prediction [14].

Despite this progress, there remains a practical systems problem. Forecasting models are frequently evaluated independently of the operational workflow in which their predictions are consumed. A warehouse or procurement user may require not only a predicted demand value but also an indication of inventory risk, an explanation of the prediction, scenario analysis, and the ability to review and override automated recommendations.

This paper addresses that systems-integration problem through a modular web-based Decision Support System (DSS). The system combines historical demand-related features with multiple supervised regression models and connects the resulting forecast to inventory calculations, scenario analysis, approximate feature-sensitivity explanations, and a human-in-the-loop approval workflow.

The principal contributions of this work are:

1) A modular web-based architecture integrating a forecasting pipeline with an inventory decision-support layer.
2) A chronological multi-model forecasting experiment comparing Linear Regression, Ridge Regression, Decision Tree, and Random Forest models.
3) A leakage-aware temporal formulation in which previous-period sales are distinguished from the target-period demand.
4) Integration of empirical forecast uncertainty and approximate feature-sensitivity information alongside point predictions.
5) A deterministic inventory-risk layer incorporating safety stock and reorder-point calculations.
6) A scenario-analysis module allowing users to evaluate changes in selected operational assumptions.
7) A human-in-the-loop approval mechanism that records procurement decisions and user justification.

The contribution is therefore positioned primarily as a decision-support and systems-integration contribution rather than as a new machine-learning algorithm.

The remainder of this paper is organized as follows. Section II reviews recent literature. Section III defines the problem and objectives. Section IV describes the system architecture. Section V presents the forecasting methodology and inventory formulation. Section VI describes the experimental setup and results. Section VII discusses the decision-support features. Section VIII presents limitations. Section IX describes future research directions. Section X concludes the paper.

II. LITERATURE REVIEW

A. Machine Learning for Demand Forecasting

Recent literature demonstrates increasing adoption of machine learning and deep learning for supply-chain demand forecasting. Douaioui *et al.* review 119 studies published between 2015 and 2024 and identify a broad range of ML and DL methods used for supply-chain forecasting [9]. Goel *et al.* investigate demand forecasting using machine-learning methods across supply-chain contexts and emphasize the importance of selecting appropriate forecasting strategies for different demand horizons [10].

Within IEEE literature, Poongothai *et al.* investigate ML-based demand forecasting and inventory optimization in retail [1]. Zheng examines the application of machine-learning algorithms to enterprise supply-chain demand forecasting [3]. Singh *et al.* compare deep-learning approaches for forecasting and inventory optimization in e-commerce supply chains [4]. These studies establish the relevance of predictive modeling to operational supply-chain decisions.

B. Inventory-Oriented Forecasting

Demand forecasting becomes more operationally useful when predictions are connected to inventory decisions. Sukolkit *et al.* compare multiple forecasting methods for steel-industry demand and explicitly discuss how forecast results can support inventory management [11]. Wang investigates intelligent demand prediction together with inventory optimization in e-commerce [2].

van der Haar *et al.* examine supervised learning for integrated forecasting and inventory control and show that predictive performance and inventory-control objectives can be considered together [12]. Nguyen *et al.* extend this perspective using advanced time-series forecasting together with probabilistic inventory models involving safety stock, reorder points, and inventory costs [15].

C. Retail and E-Commerce Applications

Retail and e-commerce environments present additional challenges because demand may change with product characteristics, pricing, market conditions, and customer behavior. Poongothai *et al.* address retail demand forecasting and inventory optimization using machine learning [1]. Wang considers intelligent demand prediction and inventory optimization in e-commerce [2]. Singh *et al.* investigate deep-learning models using sales, inventory, market, customer, and supplier information for e-commerce supply chains [4].

Sathyabama *et al.* investigate an AI-driven retail inventory system incorporating ARIMA and LSTM for expiry prediction and dynamic pricing [5]. Manos *et al.* propose a dynamic demand forecasting platform that considers multiple information sources and supply-chain value [6].

D. Interpretability and Inventory Risk

Interpretability and inventory risk are important when forecasts influence operational decisions. An *et al.* propose association-rule-based machine learning for demand forecasting in micro-fulfillment centers and use associations to improve interpretability [13]. Liu *et al.* investigate machine learning for inventory stockout prediction, highlighting the relationship between predictive analytics and inventory risk [14].

Zuvieta *et al.* survey statistical and ML-based demand forecasting methods for spare parts and emphasize that forecasting quality should be considered together with the costs associated with over- and under-forecasting [7]. Nurhasanah *et al.* demonstrate the use of neural networks for demand forecasting within a supply-chain network and connect forecasting performance to production and distribution planning [8].

E. Research Gap

The reviewed literature demonstrates substantial progress in demand forecasting, inventory optimization, and AI-based supply-chain analytics. However, many studies focus primarily on the forecasting algorithm or on a specific operational setting. A practical decision-support implementation must additionally connect predictions to inventory-risk calculations, scenario analysis, interpretability, and user approval.

The research gap addressed in this work is therefore not the absence of forecasting algorithms. Instead, it concerns the integration of multiple analytical components into a single auditable workflow suitable for inventory decision support. The present work consequently contributes a prototype architecture that combines predictive modeling, inventory calculations, uncertainty information, scenario analysis, and human review.

III. PROBLEM STATEMENT AND OBJECTIVES

Retailers and inventory managers must determine appropriate replenishment quantities despite uncertainty in future demand. Static reorder thresholds and manual estimation can become inadequate when sales patterns, inventory levels, prices, and supplier lead times change.

The problem addressed by this work is the design of a decision-support pipeline that:

1) estimates future demand from historical operational variables;
2) compares multiple supervised learning models using a chronological evaluation procedure;
3) translates forecasts into inventory-risk indicators;
4) provides uncertainty and approximate feature-sensitivity information;
5) enables scenario-based what-if analysis; and
6) retains human authority over final procurement decisions.

The specific objectives are:

1) Develop a modular web-based inventory decision-support system.
2) Train and compare Linear Regression, Ridge Regression, Decision Tree, and Random Forest models.
3) Maintain a chronological relationship between predictor variables and target-period demand.
4) Integrate serialized ML models into the application backend.
5) Report forecasting metrics and empirical uncertainty information.
6) Compute inventory-risk indicators such as Safety Stock and Reorder Point.
7) Provide scenario-based sensitivity analysis.
8) Implement a human-in-the-loop approval and audit workflow.

IV. SYSTEM ARCHITECTURE

The proposed system uses a modular architecture consisting of a frontend presentation layer, backend API layer, persistence layer, machine-learning inference layer, and inventory decision engine.

Modular architecture of the Smart Inventory Demand Forecasting Decision Support System.

A. Frontend Layer

The frontend provides the primary interaction interface for inventory managers. It displays historical inventory information, forecasts, risk indicators, scenario outputs, and pending procurement recommendations.

B. Backend API Layer

The backend is implemented using Node.js and Express. It handles authentication, application requests, data access, model invocation, and communication with the decision engine.

Authentication uses JSON Web Tokens (JWTs) for protected API requests. The authentication mechanism is intended to control access to system functions rather than to constitute a separate research contribution.

C. Persistence Layer

The prototype uses localized JSON files for development and experimental persistence. This includes historical observations, user information, procurement records, and audit information.

The JSON-based architecture is explicitly treated as a prototype implementation and not as evidence of production-scale database performance.

D. Machine Learning Layer

The forecasting pipeline is implemented in Python using `pandas` and `scikit-learn`. Trained models are serialized locally and loaded during inference.

The backend invokes the forecasting process and passes the required feature vector. The resulting forecast is returned to the decision engine.

TABLE I
COMPARISON OF RECENT RESEARCH RELATED TO DEMAND FORECASTING AND INVENTORY DECISION SUPPORT

Reference	Primary Focus	Main Contribution	Relation to Proposed System
[1]	Retail demand forecasting and inventory optimization	Applies ML to retail demand prediction and inventory planning	Supports integration of forecasting with retail inventory decisions
[2]	Deep learning for inventory optimization and demand prediction	Combines intelligent demand prediction with inventory optimization	Motivates forecast-to-inventory integration
[3]	Enterprise supply-chain demand forecasting	Evaluates machine-learning algorithms for supply-chain demand prediction	Supports comparative ML evaluation
[4]	Deep learning for e-commerce supply chains	Studies demand forecasting and inventory optimization using deep learning	Provides advanced-model direction for future extensions
[5]	Retail inventory, expiry prediction and dynamic pricing	Combines forecasting with retail inventory and pricing decisions	Supports operational decision-support perspective
[6]	Dynamic AI-based supply-chain forecasting	Combines demand forecasting with supply-chain value analysis	Motivates integrated forecasting workflows
[7]	Survey of statistical and ML demand forecasting	Reviews forecasting methods for inventory-oriented spare parts	Supports model-selection and inventory-risk discussion
[8]	Supply-chain demand forecasting using ANN	Applies ANN to multi-product supply-chain demand	Supports multi-product forecasting formulation
[9]	Review of ML/DL demand forecasting	Systematic review of ML and DL methods	Provides broad literature foundation
[11]	Inventory-oriented demand forecasting	Connects forecasting models with inventory management	Directly supports forecast-to-inventory integration
[12]	Integrated forecasting and inventory control	Uses supervised learning with inventory-control objectives	Supports decision-support formulation
[15]	Forecasting and inventory under uncertainty	Integrates advanced forecasting with safety stock and inventory cost	Supports uncertainty-aware inventory extension
[13]	Interpretable ML for demand forecasting	Uses association rules for feature selection and interpretability	Supports explainability requirements
[14]	Inventory stockout prediction	Uses ML for inventory stockout risk	Supports inventory-risk analytics

V. METHODOLOGY

A. Dataset

The experimental dataset contains 48 chronological observations representing four products across twelve months.

The products are:

- Laptop Pro X1
- Wireless Headphones
- Smart Watch S3
- Gaming Mouse

Product identity is retained for chronological organization and product-level analysis. The reported baseline model is a pooled regression model using a common five-variable predictor set. Product identity is therefore not included as one of the five numerical predictors in the baseline experiment.

The target variable is period demand.

B. Feature Definition

The feature vector is defined as:

$$X = [Sales,\ Inventory,\ Price,\ Month,\ LeadTime] \quad (1)$$

where:

- **Sales**: previous-period sales;
- **Inventory**: recorded inventory at the prediction origin;
- **Price**: unit selling price;
- **Month**: integer month representation from 1 to 12;
- **LeadTime**: supplier delivery time in days.

The target is:

$$y = Demand_t \quad (2)$$

The temporal relationship is explicitly defined such that the Sales feature corresponds to information available before the target-period demand.

This distinction is necessary to reduce the possibility of target leakage.

C. Data Quality Assessment

Before model training, the dataset is checked for:

- missing values;
- duplicate observations;
- schema inconsistencies;
- extreme numerical observations.

Potential numerical outliers are identified using the Interquartile Range:

$$IQR = Q_3 - Q_1 \quad (3)$$

An observation is flagged when:

$$x < Q_1 - 1.5IQR \quad (4)$$

or:

$$x > Q_3 + 1.5IQR \quad (5)$$

The procedure identified 12 potential price outliers in the baseline dataset. These observations were retained for the reported baseline experiment and treated as data-quality observations rather than automatically deleted samples.

D. Preprocessing

Linear Regression and Ridge Regression use standardized numerical inputs:

$$x' = \frac{x - \mu_{train}}{\sigma_{train}} \tag{6}$$

where μ_{train} and σ_{train} are calculated only from the training data.

The same training-derived parameters are applied to the holdout observations.

Tree-based models do not require feature standardization and are trained on the corresponding unscaled inputs.

The Month variable is represented numerically from 1 to 12 in the reported baseline experiment. The limitations associated with this representation are discussed later.

E. Chronological Train-Test Split

Random shuffling is avoided because future observations must not influence model training.

For each product, the first nine monthly observations are used for training and the final three observations are used for testing.

Therefore:

$$N_{train} = 4 \times 9 = 36 \tag{7}$$

and:

$$N_{test} = 4 \times 3 = 12 \tag{8}$$

The resulting 75:25 chronological split is used for all reported models.

Because only one final holdout window is available, the experiment should be interpreted as a preliminary baseline evaluation.

F. Regression Models

Four regression algorithms are evaluated.

1) Linear Regression: Linear Regression minimizes:

$$\min_w \|Xw - y\|_2^2 \tag{9}$$

2) Ridge Regression: Ridge Regression adds an L2 regularization term:

$$\min_w \|Xw - y\|_2^2 + \alpha\|w\|_2^2 \tag{10}$$

where α controls the regularization strength.

The implementation uses:

$$\alpha = 1.0 \tag{11}$$

3) Decision Tree: The Decision Tree Regressor recursively partitions the feature space and predicts values at terminal nodes.

The reported implementation constrains the maximum tree depth to five.

4) Random Forest: The Random Forest consists of multiple decision trees trained using bootstrap samples.

The final prediction is:

$$\hat{y} = \frac{1}{K}\sum_{k=1}^{K} f_k(x) \tag{12}$$

where K represents the number of trees.

The reported implementation uses:

$$K = 100 \tag{13}$$

TABLE II
MODEL CONFIGURATION

Model	Configuration
Linear Regression	`fit_intercept=True`
Ridge	$\alpha = 1.0$
Decision Tree	`max_depth=5`
Random Forest	100 trees

VI. MODEL EVALUATION

Four metrics are used.

A. Mean Absolute Error

$$MAE = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i| \tag{14}$$

B. Root Mean Squared Error

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{15}$$

C. Mean Absolute Percentage Error

$$MAPE = \frac{100}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \tag{16}$$

MAPE is interpreted cautiously because it becomes unstable when actual demand approaches zero.

D. Coefficient of Determination

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{17}$$

The primary model-selection criterion is MAE because it directly expresses forecast error in demand units.

RMSE, MAPE, and R^2 are reported as complementary measures.

VII. UNCERTAINTY AND EXPLAINABILITY

A. Empirical Uncertainty

For Random Forest, the standard deviation of individual tree predictions is used as an empirical dispersion measure.

Let:

$$\sigma_{trees} = Std(f_1(x), f_2(x), ..., f_K(x)) \tag{18}$$

An empirical normal-approximation band is:

$$U_{90} = \hat{y} \pm 1.645\sigma_{trees} \tag{19}$$

This quantity is referred to as an empirical uncertainty band rather than a statistically guaranteed 90% prediction interval.

Its coverage has not been calibrated using a sufficiently large independent evaluation set.

B. Ridge Residual Uncertainty

For Ridge Regression, uncertainty is estimated from the chronological training residuals:

$$e_i = y_i - \hat{y}_i \tag{20}$$

and:

$$\sigma_e = \sqrt{\frac{1}{n_{train} - 1} \sum_{i=1}^{n_{train}} (e_i - \bar{e})^2} \tag{21}$$

The corresponding empirical uncertainty band is:

$$U_{Ridge} = \hat{y} \pm 1.645\sigma_e \tag{22}$$

This residual-based band is also not claimed to have guaranteed statistical coverage.

C. Approximate Feature Sensitivity

The current implementation uses baseline perturbation rather than SHAP.

A baseline feature vector is first passed through the model. Individual features are subsequently replaced with their active observation values and the prediction difference is measured.

For feature j:

$$C_j = f(x^{(j)}_{baseline}) - f(x_{baseline}) \tag{23}$$

The resulting value is interpreted as an approximate model-specific sensitivity measure.

It is not claimed to provide the formal additive attribution properties of SHAP.

VIII. INVENTORY DECISION LOGIC

The forecasting output is passed to a deterministic inventory decision layer.

A. Daily Demand

For a monthly forecast $\hat{D}$:

$$\bar{d} = \frac{\hat{D}}{30} \tag{24}$$

B. Safety Stock

The generalized safety-stock formulation is:

$$SS = Z\sqrt{\bar{L}\sigma_d^2 + \bar{d}^2\sigma_L^2} \tag{25}$$

where:

- Z is the selected service-level factor;
- $\bar{L}$ is mean supplier lead time;
- σ_d is demand standard deviation;
- σ_L is lead-time standard deviation.

For the current prototype, lead-time variability is assumed to be negligible:

$$SS = Z\sigma_d\sqrt{\bar{L}} \tag{26}$$

The implementation uses:

$$Z = 1.65 \tag{27}$$

as an approximate service-level factor.

C. Reorder Point

The reorder point is:

$$ROP = \bar{d}\bar{L} + SS \tag{28}$$

D. Recommended Order Quantity

Let I denote current inventory.

The prototype recommendation is:

$$Q = \max\left(0, \hat{D} + SS - I\right) \tag{29}$$

This equation is explicitly treated as a deterministic prototype replenishment rule.

It is not claimed to be an optimal cost-minimizing inventory-control policy.

IX. EXPERIMENTAL RESULTS

A. Experimental Environment

The prototype was evaluated using a development environment containing an Intel Core i7 processor and 16 GB RAM.

The software environment consisted of:

- Node.js 18.x;
- Python 3.10;
- pandas 2.0;
- scikit-learn 1.2.

B. Forecasting Results

The four models were evaluated using the same chronological holdout set containing 12 observations.

Ridge Regression achieved the lowest MAE and MAPE.

Linear Regression achieved a lower RMSE and slightly higher R^2.

Therefore, the selection of Ridge is based specifically on the predefined MAE criterion.

The results do not demonstrate that Ridge is superior under every possible metric.

TABLE III
MODEL PERFORMANCE ON THE 12-OBSERVATION HOLDOUT SET

Model	MAE	RMSE	MAPE	R^2
Random Forest	48.10	60.30	5.97%	0.9140
Linear Regression	13.87	16.58	2.00%	0.9935
Ridge Regression	**13.39**	17.70	**1.70%**	0.9926
Decision Tree	44.72	54.34	5.58%	0.9304

C. Interpretation of High R^2

The linear models produced R^2 values above 0.99.

Such values are unusually high for a small forecasting experiment.

Several explanations are possible, including strong relationships within the supplied dataset, structured data generation, or information leakage.

The current paper does not claim to establish which explanation is responsible.

The temporal relationship between previous-period Sales and target-period Demand must be preserved during dataset construction.

The high R^2 values should therefore be interpreted cautiously, especially because the final evaluation contains only twelve observations.

D. Comparison With Literature

The observed results demonstrate that relatively simple regularized linear models can perform strongly on the supplied baseline dataset. This does not imply that they will outperform deep-learning or ensemble models on larger real-world datasets.

Recent studies have reported benefits from more advanced methods under different data conditions. Singh *et al.* investigate deep-learning approaches for e-commerce supply chains [4], while Wang examines deep learning for inventory optimization and intelligent demand prediction [2]. Nguyen *et al.* further demonstrate the potential of advanced time-series models when forecasting and inventory objectives are jointly considered [15].

The present work therefore treats the classical regression models as transparent baseline models and identifies advanced forecasting architectures as future extensions.

X. DECISION SUPPORT FEATURES

A. Scenario Analysis

The system provides a what-if scenario interface.

Users can modify selected operational assumptions, including:

- expected demand;
- selling price;
- supplier lead time;
- current inventory.

The modified feature vector is passed through the forecasting pipeline and the resulting forecast and inventory indicators are displayed without permanently modifying the original historical record.

This allows users to examine possible operational outcomes before approving a procurement action.

B. Human-in-the-Loop Approval

The system does not automatically execute procurement actions.

Recommendations are placed in a pending queue.

The user can select:

- `APPROVE`;
- `REJECT`;
- `SNOOZE`.

For rejected or deferred recommendations, the user provides a contextual justification.

The backend records:

- user identifier;
- timestamp;
- selected model;
- forecast;
- recommendation;
- user action;
- justification.

This creates an audit trail for decision-support actions.

XI. LIMITATIONS

The current study has several important limitations.

First, the dataset contains only 48 observations across four products and twelve months. The resulting 12-observation test set is too small to establish broad generalization.

Second, the experiment uses a single chronological holdout. Rolling-origin or walk-forward validation would provide stronger evidence by evaluating multiple temporal windows.

Third, Product/SKU identity is not included as an explicit predictor in the pooled baseline model. Future experiments should compare pooled models with explicit SKU encoding and product-specific models.

Fourth, Month is represented as an integer from 1 to 12. This representation does not explicitly model cyclical relationships between December and January. Future work should compare numerical, one-hot, and cyclical encodings.

Fifth, the dataset contains 12 potential price outliers identified by the IQR diagnostic. They are retained in the baseline experiment, but their influence should be examined using sensitivity analysis.

Sixth, the current implementation does not explicitly model censored demand. Historical sales can underestimate true demand when products are unavailable during part of a period.

Seventh, uncertainty estimates are empirical and are not statistically calibrated. Formal prediction-interval calibration requires larger independent evaluation datasets.

Eighth, the JSON persistence layer is appropriate for a prototype but does not establish scalability for production environments.

Ninth, the approximate feature-sensitivity method does not provide the formal attribution guarantees associated with SHAP.

Finally, the deterministic order-quantity formula is a prototype rule and does not optimize a complete cost function involving holding, ordering, shortage, and service-level costs.

XII. FUTURE RESEARCH

Future research should first obtain larger multi-year datasets containing multiple SKUs and more diverse demand patterns.

The evaluation should then be extended from a single holdout to rolling-origin or walk-forward validation.

Product identity should be explicitly evaluated as a model feature, and pooled models should be compared against product-specific models.

Future forecasting experiments can compare the current regression baselines against:

- Random Forest ensembles with tuned hyperparameters;
- Gradient Boosting;
- XGBoost;
- LSTM;
- GRU;
- Transformer-based models;
- modern time-series mixer architectures.

Advanced models should only be introduced when the dataset size supports meaningful training and evaluation.

The inventory layer can also be extended using cost-aware optimization. Instead of selecting an order quantity solely from demand and safety stock, future versions can explicitly model:

- holding cost;
- stockout cost;
- ordering cost;
- lead-time uncertainty;
- service-level constraints.

The uncertainty module can be extended using conformal prediction or calibrated probabilistic forecasting.

The explainability layer can be evaluated using SHAP or other formally defined attribution methods.

Finally, the prototype JSON persistence layer can be replaced with a relational database and evaluated under realistic multi-user workloads.

XIII. CONCLUSION

This paper presented a Smart Inventory Demand Forecasting Decision Support System that integrates machine-learning-based demand forecasting with inventory-risk calculations, scenario analysis, approximate model interpretation, and human-in-the-loop procurement review.

Four supervised regression models were evaluated on a 48-observation chronological dataset. Ridge Regression obtained the lowest MAE of 13.39 and MAPE of 1.70%, while Linear Regression achieved a slightly lower RMSE of 16.58 and a higher R^2 of 0.9935.

These results are treated as preliminary because only twelve observations were used for final holdout evaluation. The study therefore does not claim production-level forecasting accuracy or generalization to broader commercial datasets.

The primary contribution is the integration of forecasting and inventory decision support into an auditable workflow. The system connects demand predictions with safety-stock and reorder-point calculations, scenario analysis, empirical uncertainty information, approximate feature sensitivity, and human approval.

The literature demonstrates that more advanced forecasting and inventory-optimization approaches are increasingly being explored, including deep-learning, interpretable ML, and uncertainty-aware inventory models. The proposed system provides a transparent baseline architecture from which such approaches can be evaluated in future work.

Larger datasets, rolling-origin validation, explicit SKU modeling, calibrated uncertainty estimates, and cost-aware inventory optimization are required before the system can be considered for production deployment.

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